

# **Audio ML: Urban Soundscape Classification**

## **Abstract**

The Audio ML: Urban Soundscape Classification project aims to develop an intelligent system capable of recognising and categorising various urban environmental sounds using machine learning and deep learning techniques. These sounds include everyday noises such as sirens, car horns, drilling, street music, and dog barks, which collectively form the soundscape of urban environments.

The project focuses on automating the identification and classification of these sounds to support applications like noise pollution monitoring, smart city planning, and urban event detection. Manual sound analysis is often time-consuming and inconsistent, whereas an ML-based approach ensures greater speed, accuracy, and scalability.

The workflow involves data collection (using datasets like UrbanSound8K), preprocessing and augmentation using Librosa, PyDub, SoundFile, and SciPy, followed by feature extraction of acoustic properties such as MFCCs and Spectrograms. Models are then built using Keras or PyTorch frameworks, trained and evaluated on categorized datasets, and finally deployed for real-time sound recognition or urban noise analytics.

This project provides a robust framework for analysing city sound environments, offering insights into urban noise behaviour and enabling the development of smarter, safer, and more sustainable cities through sound-based monitoring technologies.

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